

PROJECT SYNOPSIS

SUPERSTORE SALES OVERVIEW & GEOGRAPHIC INSIGHTS

End-to-End Cloud Analytics using AWS, Amazon Athena and Power BI

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Project Theme :

The project presents an end-to-end cloud-based analytics workflow in which a raw Superstore dataset is stored in Amazon S3, catalogued through AWS Glue, queried using Amazon Athena, and connected to Power BI through an Athena ODBC connection. The prepared data is transformed into interactive dashboards for sales, profitability, geographic and customer-oriented analysis

Synopsis at a Glance :

- Cloud storage and organization of the raw dataset using Amazon S3.
- Secure resource access through AWS IAM and a service role for AWS Glue.
- Automated schema discovery and metadata management using AWS Glue Crawler and Data Catalog.
- Serverless SQL-based analysis using Amazon Athena with filtering, aggregation, grouping, counts, averages and threshold analysis.
- Data cleaning and transformation in Power Query, followed by dynamic KPI calculations using DAX.
- Interactive two-page Power BI dashboard covering Sales Overview and Profitability Analysis.

Technology Stack :

- Amazon S3 • AWS IAM • AWS Glue • Amazon Athena • ODBC • Power BI • Power Query • DAX

1. INTRODUCTION

- Modern organizations generate large volumes of operational data, but raw data becomes useful only after it is securely stored, structured, cleaned, analyzed and communicated through meaningful visualizations.
- This project demonstrates a practical cloud analytics pipeline using the Superstore dataset. The workflow combines AWS cloud services with SQL analysis and Power BI visualization to convert raw records into business-oriented insights.
- The project follows a connected flow from data storage to visualization. Amazon S3 acts as the cloud storage layer, AWS IAM provides controlled access, AWS Glue discovers and catalogs the dataset schema, and Amazon Athena provides a serverless SQL environment for analysis.
- The resulting data is connected to Power BI through an ODBC connection, where Power Query performs data preparation and DAX measures provide dynamic business calculations

2. PROJECT OBJECTIVES

- Develop an end-to-end cloud-based data analytics solution using the Superstore dataset.
- Store and manage raw data in Amazon S3 using separate source and query-result storage locations.
- Configure secure and controlled access to AWS resources using IAM.
- Automatically identify data structure and create catalog metadata using AWS Glue Crawler and Data Catalog.
- Perform SQL-based filtering, aggregation, grouping, counting, averaging and threshold analysis using Amazon Athena.
- Connect the cataloged dataset to Power BI through the Simba Athena ODBC driver. • Clean, transform and standardize data using Power Query.
- Create DAX measures for sales, profit, orders, quantity, discount and profitability KPIs.
- Build an interactive dashboard that supports business analysis and data-driven decision-making.

3. PROBLEM STATEMENT

Raw business datasets often contain inconsistent formats, mixed data types and large numbers of records that are difficult to interpret directly. A structured analytical solution is therefore required to organize the data, provide secure access, support SQL-based exploration and present the resulting information in an interactive format. This project addresses that requirement by integrating AWS data services with Power BI

4. SCOPE OF THE PROJECT

- Cloud-based storage, cataloging and querying of the Superstore dataset.
- SQL analysis at region, category and shipping-mode levels.
- Preparation of analytical data for visualization.
- Creation of KPI-driven Power BI dashboards for management-oriented interpretation.
- Use of reusable cloud and BI components that can support further analysis.

5. METHODOLOGY / SYSTEM WORKFLOW

The methodology is organized as a sequence of data engineering, data analysis and visualization stages. Each stage prepares the output required by the next stage.

Stage	Technology	Main Activity	Output
1	Amazon S3	Upload and store raw Superstore CSV data.	Cloud-based source data
2	AWS IAM	Provide permissions required for Glue access.	Controlled access role
3	AWS Glue	Create database, crawler and catalog metadata.	Cataloged table/schema
4	Amazon Athena	Run SQL queries on the cataloged table.	Analytical query results
5	Athena ODBC	Expose Athena data to Power BI.	BI connectivity
6	Power Query	Clean, transform and format data.	Analysis-ready dataset
7	DAX / Power BI	Create KPIs and interactive visuals.	Business dashboard

6. AWS CLOUD DATA INFRASTRUCTURE

- Amazon S3: Two buckets are used in the documented workflow: a source bucket for the raw dataset and a results bucket for Athena query outputs. The Superstore dataset is uploaded as a CSV file for subsequent processing.
- AWS IAM: An IAM role is configured with permissions required for the Glue service to access S3 and write metadata to the Glue Data Catalog. The documented permissions include AmazonS3FullAccess and AWSGlueServiceRole.
- AWS Glue: A Glue database is created to hold catalog metadata. A crawler is configured against the S3 data source and run on demand. After execution, the crawler creates and verifies the table in the Glue Data Catalog.
- Amazon Athena: Athena acts as the serverless SQL layer. The Glue Data Catalog database and table are selected, an S3 query-result location is configured, and SQL statements are executed directly against the cataloged data.

7. SQL ANALYSIS

- Filtering records using WHERE conditions, including region and shipping-mode analysis.
- Using MIN, MAX, AVG and SUM for sales and profit analysis.
- Using GROUP BY for region-wise and category-wise sales and profit.
- Counting records by region and shipping mode.
- Calculating average sales and average profit by category.
- Identifying products with sales greater than a specified threshold.

8. POWER BI DATA PREPARATION AND VISUALIZATION

- After SQL analysis, the Athena table is connected to Power BI Desktop through the configured Simba Amazon Athena ODBC driver. The ODBC connection enables Power BI to query the Athena table directly.
- The Superstore data is loaded into Power BI and prepared using Power Query.
- Power Query Transformation: The documented preparation process handles null values, mixed date formats and data types.
- These transformations improve data quality and consistency before analytical calculations and visualization are performed.
- DAX Measures and Calculated Logic: DAX is used to create dynamic measures for Total Sales, Total Profit, Profit Margin, Loss Orders and other business KPIs. Additional measures include Profit Margin %, Average Order Value and Average Quantity per Order. The measures respond dynamically to selected filters in the dashboard.

9. DASHBOARD DESIGN

Dashboard Area	Purpose	Representative Analysis
Sales Overview	High-level view of sales performance.	Sales, profit, orders, quantity, discount and KPI-based analysis
Profitability Analysis	Detailed view of profitability and business performance.	Profitability measures, comparative visuals and analytical breakdowns
Geographic / Customer Insights	Location and customer-oriented interpretation documented in the project objective.	Geographic visuals, customer analysis, charts and informative elements

10. Dashboard Overview

The Power BI dashboard is divided into two pages:

Page 1 – Sales Overview

- Provides an overall view of the sales performance.
- Displays important KPIs such as **Total Sales, Total Profit, Orders, Quantity, and Discount**.
- Presents sales and profit performance through interactive charts and visualizations.
- Provides **region-wise and category-wise** sales analysis.
- Uses interactive **filters and slicers** for dynamic data exploration.
- Helps identify important sales trends and performance patterns.

Page 2 – Profitability Analysis

- Focuses on analyzing the overall **profitability of the business**.
- Displays profitability-related KPIs and analytical measures.
- Includes **Profit Margin %, Average Order Value, and Average Quantity per Order**.
- Provides profit analysis across different **regions and categories**.
- Uses charts and tables to compare profitability performance.
- Helps identify **profitable and less-profitable areas**.
- Supports better **business analysis and data-driven decision-making**.

10. KEY ANALYTICAL CAPABILITIES

- Region-wise comparison of total sales and profit. Profitability measures, comparative visuals and analytical breakdowns
- Category-wise comparison of sales, profit and average values.
- Order-count analysis by region and shipping mode.
- Threshold-based identification of high-sales products.
- Dynamic KPI calculations that change according to selected dashboard filters.
- Interactive charts, KPI cards, maps and tables for easier interpretation.

11. TECHNOLOGIES USED

Technology	Role in the Project
Amazon S3	Scalable cloud storage for raw data and query results.
AWS IAM	Identity and role-based access control.
AWS Glue	Database, crawler, schema discovery and Data Catalog.
Amazon Athena	Serverless SQL query execution and analytical exploration.
Athena ODBC	Connectivity layer between Athena and Power BI.
Power Query	Data cleaning, transformation and formatting.
DAX	Dynamic measures and KPI calculations.
Power BI	Interactive dashboard and business visualization.

12. EXPECTED / DOCUMENTED OUTCOME

- The completed project demonstrates a connected cloud analytics workflow in which raw Superstore data is moved from storage to cataloging, SQL analysis, preparation and interactive visualization.
- The outcome is a structured analytical environment that allows users to explore sales and profitability information through queries and Power BI dashboards.
- The documented solution emphasizes the integration of cloud data infrastructure and business intelligence rather than treating data storage, analysis and visualization as separate activities.
- The combination of AWS Glue and Athena provides a cataloged and queryable data layer, while Power Query and DAX prepare the data for dynamic dashboard-based analysis.

13. ADVANTAGES OF THE PROPOSED SOLUTION

- Provides an end-to-end workflow from raw data storage to business visualization.
- Uses scalable cloud storage through Amazon S3.
- Automates schema discovery and cataloging through AWS Glue Crawler.
- Supports serverless SQL analysis using Amazon Athena.
- Enables controlled access through IAM roles and permissions.
- Improves data quality through Power Query transformations.
- Provides dynamic and interactive KPIs through DAX.
- Makes sales and profitability analysis easier to interpret through Power BI visuals.
- Creates a reusable foundation for further analytical and business intelligence tasks.

14. CONCLUSION

- The Superstore Sales Overview & Geographic Insights project successfully demonstrates an end-to-end cloud-based analytics approach using AWS services and Power BI. The raw Superstore dataset is stored in Amazon S3, accessed through controlled IAM permissions, automatically cataloged using AWS Glue, and analyzed through Amazon Athena SQL queries.
- The data is then connected to Power BI through ODBC, cleaned using Power Query and enriched with DAX-based KPI calculations. The final dashboard-oriented workflow transforms raw transactional data into structured, interactive information for sales, profit, order, geographic and customer-oriented analysis. Overall, the project provides practical exposure to cloud data engineering, SQL analytics, data preparation and business intelligence visualization, while supporting data-driven decision-making.

15. FUTURE ENHANCEMENTS

- Extend the dashboard with additional time-based trend analysis.
- Add more advanced customer segmentation and profitability analysis.
- Introduce automated refresh and monitoring for regularly updated datasets.
- Expand the analytical model with additional business KPIs and comparative measures.
- Integrate additional datasets to support broader business analysis.